

Backup and Recovery Integration Documentation

GrocerEase Inventory Management System

1. Integration Summary

The backup and recovery module was rebuilt around a reusable BackupManager class. The controller, Settings page, Dashboard quick actions, and scheduled backup script now use the same backup implementation for consistent behavior.

2. Files Integrated

- models/BackupManager.php: creates backups, validates backup files, lists history, and restores SQL.
- controllers/backup.php: exposes authenticated JSON actions for backup, restore, validation, listing, and download.
- views/inventory/settings.php: provides backup controls, restore validation, backup history, and SweetAlert feedback.
- views/inventory/dashboard.php: adds quick actions for backup, recovery, products, stock, and logs.
- cron_backup.php: uses BackupManager for automatic backups.
- autoload.php: supports CLI execution for scheduled backups.

3. Backup Flowchart

```
[Admin clicks Backup Now]
|
[SweetAlert confirmation]
|-- cancel --> [No change]
|-- confirm -> [POST controllers/backup.php?action=backup]
|
[BackupManager reads tables]
|
[Write CREATE TABLE SQL]
|
[Write INSERT data SQL]
|
[Validate .sql file]
|-- fail --> [Show error alert]
|-- pass --> [Save in /backups]
|
[Refresh history]
```

4. Recovery Flowchart

```
[Admin selects backup]
|
[Validate file extension, readability, size, SQL contents]
|-- invalid --> [Show validation error]
|-- valid ----> [SweetAlert restore confirmation]
```

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```
|
[Create pre_restore safety backup]
|
[Disable foreign key checks]
|
[Execute SQL statements]
|
[Enable foreign key checks]
|-- error --> [Show error + safety backup name]
|-- success -> [Show success + refresh history]
```

5. Validation Added

- User must be logged in before backup controller actions run.
- Only POST can create or restore backups; only GET can list or download.
- Restore files must be .sql, readable, non-empty, and SQL-like.
- Stored backup filenames are sanitized and resolved inside the backups folder only.
- Uploaded files are checked with `is_uploaded_file` before restore.
- A safety backup is created before every restore attempt.

6. Error Handling

All backup controller actions return JSON with a status and message. The interface now checks `data.status` instead of `data.success`, so SweetAlert success and error messages match the actual controller response.

7. Demonstration Notes

- Use Create Backup before changing records.
- Modify or archive a visible record, then show that the data changed.
- Use Recovery Module to restore the selected backup.
- Show restored products, suppliers, stock, or logs to prove effectiveness.